



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-01 Targeted Remediation Cards

Repair only the weak ingestion competency

Artifact	C3_01_TARGETED_REMEDIATION_CARDS_RC1
Status	2026 - LEARNER EDITION
Teaching standard	M01 beginner-first teacher loop. Videos are visual companions; the book remains sufficient fallback.
C2B boundary	Bridge foundations may be assumed, but C3 engineering behavior is demonstrated, broken, repaired, transferred and validated.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

Course rule: preserve what arrived -> validate interpretation -> quarantine ambiguity -> publish durably -> commit state last -> prove replay safety. A green job is never sufficient evidence by itself.



G1 repair - Source contract & landing

PASS evidence expected: Inputs/windows/grains declared; raw + manifest/fingerprint preserved.

REPAIR LOOP

1) name the exact failed invariant; 2) reopen only the matching lesson; 3) follow one tiny fresh example; 4) close all material; 5) solve a changed micro-case; 6) run an independent control; 7) explain the prior failure and new recovery; 8) save as append-only repair evidence.

DO NOT

Do not restart the whole module, overwrite Gate attempt_001, average this competency with stronger ones, or open Gate B/C without explicit assignment.

RETURN WHEN

Changed micro-case proves this competency independently and no related critical failure remains.



G2 repair - Defensive parsing & evolution

PASS evidence expected: CSV/JSON/schema/time behavior explicit; unsafe ambiguity quarantined/blocked.

REPAIR LOOP

1) name the exact failed invariant; 2) reopen only the matching lesson; 3) follow one tiny fresh example; 4) close all material; 5) solve a changed micro-case; 6) run an independent control; 7) explain the prior failure and new recovery; 8) save as append-only repair evidence.

DO NOT

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RETURN WHEN

Changed micro-case proves this competency independently and no related critical failure remains.



G3 repair - Incremental/CDC semantics

PASS evidence expected: Stable key/cursor/event semantics; ties, updates/deletes handled deliberately.

REPAIR LOOP

1) name the exact failed invariant; 2) reopen only the matching lesson; 3) follow one tiny fresh example; 4) close all material; 5) solve a changed micro-case; 6) run an independent control; 7) explain the prior failure and new recovery; 8) save as append-only repair evidence.

DO NOT

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RETURN WHEN

Changed micro-case proves this competency independently and no related critical failure remains.



G4 repair - State commit & replay safety

PASS evidence expected: Durable publish/validate before state; injected failure + restart proven.

REPAIR LOOP

1) name the exact failed invariant; 2) reopen only the matching lesson; 3) follow one tiny fresh example; 4) close all material; 5) solve a changed micro-case; 6) run an independent control; 7) explain the prior failure and new recovery; 8) save as append-only repair evidence.

DO NOT

Do not restart the whole module, overwrite Gate attempt_001, average this competency with stronger ones, or open Gate B/C without explicit assignment.

RETURN WHEN

Changed micro-case proves this competency independently and no related critical failure remains.



G5 repair - Interface reliability

PASS evidence expected: Auth secret boundary if relevant; timeout/pagination/retry/rate-limit rules justified.

REPAIR LOOP

1) name the exact failed invariant; 2) reopen only the matching lesson; 3) follow one tiny fresh example; 4) close all material; 5) solve a changed micro-case; 6) run an independent control; 7) explain the prior failure and new recovery; 8) save as append-only repair evidence.

DO NOT

Do not restart the whole module, overwrite Gate attempt_001, average this competency with stronger ones, or open Gate B/C without explicit assignment.

RETURN WHEN

Changed micro-case proves this competency independently and no related critical failure remains.



G6 repair - Testing/reconciliation

PASS evidence expected: Independent controls + edge/failure test; source/output/state reconciled.

REPAIR LOOP

1) name the exact failed invariant; 2) reopen only the matching lesson; 3) follow one tiny fresh example; 4) close all material; 5) solve a changed micro-case; 6) run an independent control; 7) explain the prior failure and new recovery; 8) save as append-only repair evidence.

DO NOT

Do not restart the whole module, overwrite Gate attempt_001, average this competency with stronger ones, or open Gate B/C without explicit assignment.

RETURN WHEN

Changed micro-case proves this competency independently and no related critical failure remains.



G7 repair - Operational handoff

PASS evidence expected: Sanitized logs, runbook, recovery/backfill policy, interview defense.

REPAIR LOOP

1) name the exact failed invariant; 2) reopen only the matching lesson; 3) follow one tiny fresh example; 4) close all material; 5) solve a changed micro-case; 6) run an independent control; 7) explain the prior failure and new recovery; 8) save as append-only repair evidence.

DO NOT

Do not restart the whole module, overwrite Gate attempt_001, average this competency with stronger ones, or open Gate B/C without explicit assignment.

RETURN WHEN

Changed micro-case proves this competency independently and no related critical failure remains.